

Semester One Project: What Characteristics Are Associated with Depression and Anxiety?

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Type

- Our project was a survey
- Our information was collected from students at the school
- Since categorical information was collected by directly asking, we needed to survey the people (we could not have used an experiment or observational study)
- We included two professionally made surveys instead of having to develop our own way to quantify depression and anxiety
- If there was any response bias because of how these questions are worded, it is what medical professionals have agreed is acceptable to be present within official diagnostic material
- Sampling Frame: list of Wachusett students
- Population: Wachusett students

Survey - Randomization

- Obtained a list of all Wachusett students
 - Stratified all the students into grade levels (9th, 10th, 11th, and 12th)
 - Assigned each student a unique number
 - Used a random number generator to select 50 students, with no repeats and numbers over the greatest student number
 - Repeated for each grade (4 times total)
-
- Sent emails to each selected student's school account (email included reasons for study and google form)
 - Information was recorded

The 416 members of the 2028 class roster were each assigned a number alphabetically by last name. We generated 50 random numbers, $1 \leq x \leq 416$, with no repeats:	SAMPLEING	1 10 14 27 29 31 35 47 56 62 77 82 87 99 114 119 120 123 132 155 169 177 185 194 208 209 228 232 243 262 272 285 287 289 293 312 319 320 322 328 332 366 369 377 381 383 401 403 404 414
The 425 members of the 2027 class roster were each assigned a number alphabetically by last name. We generated 50 random numbers, $1 \leq x \leq 425$, with no repeats:		13 26 33 55 68 83 92 115 131 136 154 162 173 174 177 190 193 196 203 215 220 223 225 251 253 255 259 260 264 265 279 288 292 297 298 317 336 337 340 348 355 358 387 393 395 407 409 419 420 422
The 447 members of the 2026 class roster were each assigned a number alphabetically by last name. We generated 50 random numbers, $1 \leq x \leq 447$, with no repeats:	PROCEDURE	2 18 23 28 43 53 56 70 72 80 82 83 87 95 96 99 116 151 162 164 198 202 205 206 210 226 234 239 247 256 261 273 287 289 293 306 309 327 345 347 350 353 361 382 387 403 415 429 432 444
The 475 members of the 2025 class roster were each assigned a number alphabetically by last name. We generated 50 random numbers, $1 \leq x \leq 475$, with no repeats:		19 20 29 32 34 36 43 67 81 95 101 104 105 110 127 139 144 163 177 178 180 200 219 224 227 232 253 271 282 285 299 306 328 330 351 359 364 378 379 380 381 385 394 409 429 452 453 460 473 475

	A	B	C
1	Kylie	Abbasia	Selected
2	Isabel	Abdella	
3	Afua	Adjei	
4	Theodore	Agbay	
5	Logan	Allaire	
6	Ellie	Allard	
7	Andrew	Allen	
8	Joseph	Alteri	
9	Owen	Ambach	
10	Analei	Anderson	Selected
11	Cooper	Andrews	
12	Joshua	Anom	
13	Rachel	Anom	
14	Phelipe	Apolinario Utida	Selected
15	Nirelis	Aponte-Garcia	
16	Emma	Appleby	
17	Alexis	Armstrong	
18	Simon	Atanian	
19	Dhimitra	Avushka	
20	Ryan	Ayik	
21	Maeve	Barbadora	
22	Avery	Barnes	
23	Eloisa	Barraza	
24	Alaina	Barry	
25	Owen	Barry	
26	Alexander	Barton	
27	Emily	Barton	Selected
28	Ruby	Bavosi	
29	Megan	Baylis	Selected
30	Brooke	Beardmore	
31	Chloe	Berneron	Selected

Using our randomized number lists, we selected 50 students from each grade level and manually emailed them our survey.

We decided to stratify by grade level to eliminate bias because we thought that grade level would have a large influence on some of the other questions we are asking. To avoid having grade level become a lurking variable, we wanted to equally represent all grade levels.

AP Statistics Survey



Silas Sharon <silassharon25@wrsdk12.net>

Wed, Dec 11, 2024, 12:37 PM



to bcc: Kylie, bcc: Analei, bcc: Phelipe, bcc: Emily, bcc: Megan, bcc: Chloe, bcc: Tyler, bcc: Lucas, bcc: Benjamin, bcc: Yezenia, bcc: ...

If you are receiving this email you have been selected to partake in a survey run by AP Statistics students as a midterm project. Trial responses took us 2-3 minutes. This survey seeks to identify associations between certain characteristics (e.g. sport team membership, music taste) and the symptoms of depression/anxiety. Participants were randomly selected. All responses are anonymous. All selected participants should complete the survey, regardless of any prior mental health issues or lack thereof. Your cooperation is greatly appreciated, thank you in advance!

https://docs.google.com/forms/d/e/1FAIpQLSdwZikg8-wiJF1rsrlegUJn9hGtgE7e_jBN0m2bF_tJFml9A/viewform?usp=sf_link

Questions Asked

- Grade
- Gender
- Daily Caffeine Intake
- Sport Teams Membership
- Music Genres Regularly Listened To
- Patient Health Questionnaire-9
 - Have you ever received treatment for any form of depression?
- General Anxiety Disorder-7
 - Have you ever received treatment for any form of anxiety?

Depression:

The mean depression severity is 8.144736842.

For visualization, this will be a red line.

The standard deviation is 6.62074878.

Anxiety:

The mean anxiety severity is 8.092105263.

For visualization, this will be a blue line.

The standard deviation is 5.527866324.

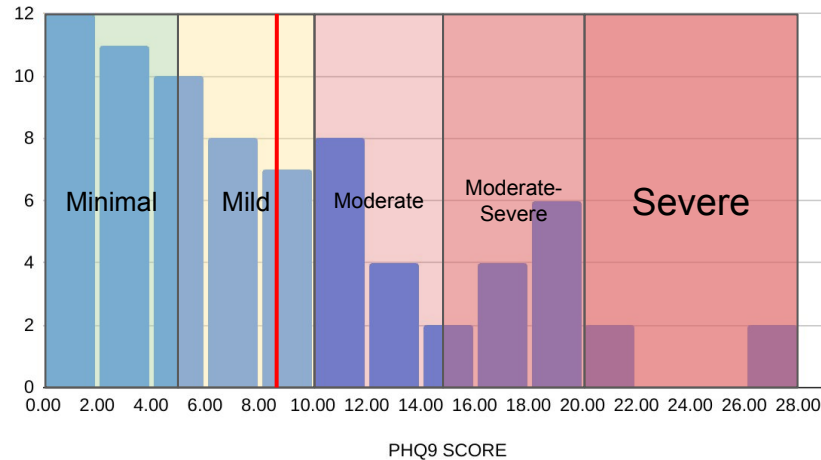


POST SCORE	PREP SCORE
1	1
12	16
8	10
5	10
5	8
8	10
14	17
10	1
8	8
14	14
6	10
8	10
14	11
21	27
10	14
10	10
6	10
17	14
8	7
12	8
10	11
10	10
14	14
14	14
8	11
8	10
10	10
10	10
10	14
12	11
8	8
1	10
6	10
8	11
1	10
20	20
1	11
10	10
8	14
8	10
14	11
12	10
14	14
6	10
11	8
8	10
8	7
8	8
8	10
14	14
10	10
11	1
10	1
6	7
8	10
8	10
8.092105263	8.144736842
5.527866324	6.62074878

HOW DEPRESSED/ANXIOUS IS THE WRHS STUDENT BODY IN GENERAL?

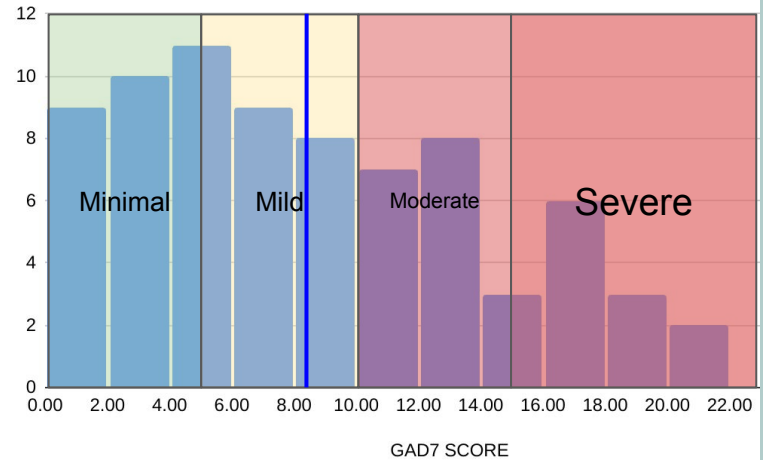
On average, a few bad weeks away from needing treatment.

Histogram of PHQ9 SCORE



Mean: 8.144
Standard Deviation: 5.528

Histogram of GAD7 SCORE



Mean: 8.092
Standard Deviation: 6.621

What is your grade level? *

- ☐ I am in 9th grade.
- ☐ I am in 10th grade.
- ☐ I am in 11th grade.
- ☐ I am in 12th grade.

DEMOGRAPHIC QUESTIONS

What is your gender? *

- ☐ I consider myself a male.
- ☐ I consider myself a female.
- ☐ I consider myself something else.

We wanted to see if being in certain grades is associated with having more severe depression/anxiety than being in other grades.

We wanted to see if there is any association between gender identity and depression/anxiety severity.

GAD Scores	Freshman	Sophomore	Junior	Senior	PHQ Scores	Freshman	Sophomore	Junior	Senior
GAD Scores	1	4	17	13	PHQ Scores	6	4	1	6
GAD Scores	18	1	13	10	PHQ Scores	16	3	19	11
GAD Scores	16	1	5	6	PHQ Scores	11	0	3	3
GAD Scores	12	3	3	4	PHQ Scores	4	8	6	7
GAD Scores	16	3	10	6	PHQ Scores	18	3	14	5
GAD Scores	15	11	8	6	PHQ Scores	18	0	11	7
GAD Scores	9	7	19	5	PHQ Scores	9	9	18	6
GAD Scores	2	16	1	12	PHQ Scores	0	16	3	8
GAD Scores	6	8	8	3	PHQ Scores	2	12	11	4
GAD Scores	14	12	4	9	PHQ Scores	11	8	16	9
GAD Scores	5	8	4	17	PHQ Scores	21	0	3	16
GAD Scores	6	16	3	12	PHQ Scores	7	27	10	14
GAD Scores	11	21	8	1	PHQ Scores	1	27	4	0
GAD Scores	6	10	7	11	PHQ Scores	4	6	11	8
GAD Scores	5	4	12	15	PHQ Scores	11	2	12	19
GAD Scores	2	18			PHQ Scores	5	18		
GAD Scores	5	0			PHQ Scores	3	1		
GAD Scores	0	3			PHQ Scores	1	4		
GAD Scores	8	10			PHQ Scores	4	13		
GAD Scores	0	20			PHQ Scores	0	20		
GAD Scores	3	0			PHQ Scores	4	0		
GAD Scores	3	5			PHQ Scores	3	2		
GAD Scores		6			PHQ Scores		0		
GAD Scores		13			PHQ Scores		12		
Mean	7.409	8.333	8.133	8.667	Mean	7.227	8.125	9.467	8.200
St. Dev.	5.646	6.604	5.263	4.670	St. Dev.	6.317	8.543	5.854	5.031

GRADE LEVEL

Depression (**Mean, StDev**):

1. Juniors (**9.467**, 5.854)
2. Seniors (**8.200**, 5.031)
3. Sophomores (8.125, 8.543)
4. Freshmen (7.227, 6.317)

Anxiety (**Mean, StDev**):

1. Seniors (**8.667**, 4.670)
2. Sophomores (8.333, 6.604)
3. Juniors (8.133, 5.263)
4. Freshmen (**7.409**, 5.646)

Depression (Mean, StDev):

1. Female (8.729, 7.082)
2. Male (7.143, 5.867)

Anxiety (Mean, StDev):

1. Female (8.646, 5.674)
2. Male (7.143, 5.338)

GENDER

GAD Scores	Male	Female	PhoQ Scores	Male	Female
GAD Scores	6	3	PhoQ Scores	8	6
GAD Scores	5	18	PhoQ Scores	11	18
GAD Scores	2	18	PhoQ Scores	5	11
GAD Scores	5	11	PhoQ Scores	3	4
GAD Scores	8	18	PhoQ Scores	1	18
GAD Scores	8	15	PhoQ Scores	4	18
GAD Scores	8	8	PhoQ Scores	8	9
GAD Scores	3	2	PhoQ Scores	4	8
GAD Scores	3	8	PhoQ Scores	3	2
GAD Scores	18	14	PhoQ Scores	11	11
GAD Scores	28	5	PhoQ Scores	28	21
GAD Scores	8	8	PhoQ Scores	8	7
GAD Scores	5	11	PhoQ Scores	2	1
GAD Scores	9	4	PhoQ Scores	8	4
GAD Scores	11	1	PhoQ Scores	11	3
GAD Scores	4	1	PhoQ Scores	3	8
GAD Scores	3	3	PhoQ Scores	18	9
GAD Scores	8	3	PhoQ Scores	4	3
GAD Scores	7	11	PhoQ Scores	11	8
GAD Scores	11	7	PhoQ Scores	11	8
GAD Scores	11	18	PhoQ Scores	8	18
GAD Scores	3	8	PhoQ Scores	4	11
GAD Scores	8	11	PhoQ Scores	8	8
GAD Scores	17	8	PhoQ Scores	18	8
GAD Scores	11	18	PhoQ Scores	14	27
GAD Scores	1	21	PhoQ Scores	8	27
GAD Scores	11	18	PhoQ Scores	8	6
GAD Scores	11	4	PhoQ Scores	18	2
GAD Scores		18	PhoQ Scores		18
GAD Scores		8	PhoQ Scores		1
GAD Scores		3	PhoQ Scores		4
GAD Scores		17	PhoQ Scores		1
GAD Scores		11	PhoQ Scores		18
GAD Scores		8	PhoQ Scores		3
GAD Scores		3	PhoQ Scores		6
GAD Scores		18	PhoQ Scores		14
GAD Scores		8	PhoQ Scores		11
GAD Scores		18	PhoQ Scores		18
GAD Scores		1	PhoQ Scores		3
GAD Scores		8	PhoQ Scores		11
GAD Scores		4	PhoQ Scores		18
GAD Scores		11	PhoQ Scores		6
GAD Scores		18	PhoQ Scores		11
GAD Scores		6	PhoQ Scores		3
GAD Scores		4	PhoQ Scores		7
GAD Scores		8	PhoQ Scores		5
GAD Scores		8	PhoQ Scores		7
GAD Scores		5	PhoQ Scores		6
Mean	7.143	8.646	Mean	7.143	8.729
St. Dev.	5.338	5.674	St. Dev.	5.867	7.082

What is your daily caffeine intake? *

- ☐ I do not consume caffeine.
- ☐ I usually have one cup of coffee or tea.
- ☐ I usually have more than one cup of coffee or tea.
- ☐ I usually have one energy drink.
- ☐ I usually have more than one energy drink.

CAFFEINE QUESTION

We wanted to see if increased caffeine consumption is associated with significantly higher rates of depression or anxiety.

This question could also illustrate a difference in association between the mental health of tea/coffee drinkers and those who drink energy drinks.

Note: If someone reported that every day they drink a beverage that contains caffeine AND that they do not consume any caffeine, we will assume that they do consume caffeine.

Depression (Mean, StDev):

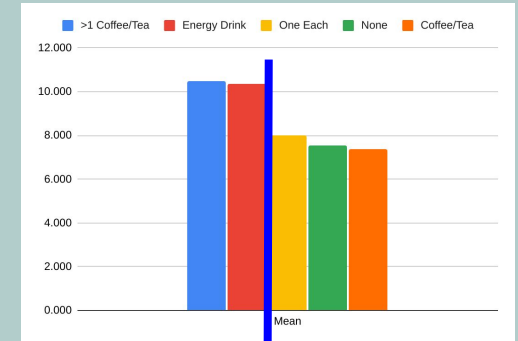
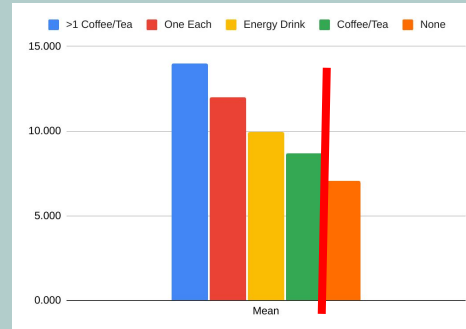
1. >1 Coffee/Tea (**14.000**, 5.657)
2. Both (**12.000**, 0.000)
3. Energy Drink (**9.929**, 7.956)
4. Coffee/Tea (**8.706**, 5.966)
5. None (**7.070**, 6.434)

Anxiety (Mean, StDev):

1. >1 Coffee/Tea (**10.500**, 10.607)
2. Energy Drink (**10.357**, 5.123)
3. Both (**8.000**, 0.000)
4. None (**7.535**, 5.692)
5. Coffee/Tea (**7.353**, 5.012)

CAFFEINE

	A	B	C	D	E	F	G	H	I	J	K	L
	GAD Score	Name	Coffee/Tea	>1 Coffee/Tea	Energy Drink	One Each	PHQ Score	Name	Coffee/Tea	>1 Coffee/Tea	Energy Drink	One Each
1	GAD Score						PHQ Score					
2	GAD Score	1	16	18	18	8	PHQ Score	6	11	18	16	12
3	GAD Score	12	15	3	8	PHQ Score	4	18	10	6		
4	GAD Score	16	5		14	PHQ Score	18	21	11			
5	GAD Score	2	1		11	PHQ Score	0	3	1			
6	GAD Score	6	2		18	PHQ Score	2	6	27			
7	GAD Score	6	3		10	PHQ Score	7	3	6			
8	GAD Score	6	3		10	PHQ Score	4	3	14			
9	GAD Score	1	17		8	PHQ Score	0	1	11			
10	GAD Score	11	4		5	PHQ Score	0	18	6			
11	GAD Score	7	6		9	PHQ Score	9	3	3			
12	GAD Score	18	4		6	PHQ Score	18	7	0			
13	GAD Score	17	6		17	PHQ Score	6	7	16			
14	GAD Score	8	8		1	PHQ Score	0	12	0			
15	GAD Score	21	4		15	PHQ Score	27	3	18			
16	GAD Score	4	7			PHQ Score	2	11				
17	GAD Score	0	12			PHQ Score	3	14				
18	GAD Score	13	11			PHQ Score	18	8				
19	GAD Score	5				PHQ Score	3					
20	GAD Score	3				PHQ Score	6					
21	GAD Score	8				PHQ Score	11					
22	GAD Score	18				PHQ Score	18					
23	GAD Score	1				PHQ Score	3					
24	GAD Score	13				PHQ Score	6					
25	GAD Score	10				PHQ Score	11					
26	GAD Score	4				PHQ Score	7					
27	GAD Score	6				PHQ Score	4					
28	GAD Score	5				PHQ Score	11					
29	GAD Score	2				PHQ Score	5					
30	GAD Score	0				PHQ Score	1					
31	GAD Score	8				PHQ Score	4					
32	GAD Score	0				PHQ Score	0					
33	GAD Score	0				PHQ Score	4					
34	GAD Score	3				PHQ Score	4					
35	GAD Score	10				PHQ Score	13					
36	GAD Score	20				PHQ Score	20					
37	GAD Score	0				PHQ Score	6					
38	GAD Score	5				PHQ Score	2					
39	GAD Score	13				PHQ Score	12					
40	GAD Score	8				PHQ Score	4					
41	GAD Score	12				PHQ Score	12					
42	GAD Score	11				PHQ Score	8					
43	GAD Score	3				PHQ Score	4					
44	GAD Score	8				PHQ Score	8					
45	Mean	7.535	7.353	10.500	10.357	8.000	Mean	7.06767142	8.705882352	14	9.929214286	12
46	St Dev	5.692	5.966	10.607	5.123	0.000	St Dev	6.43439566	5.96623308	5.65654348	7.95627611	0



SPORT TEAM QUESTION

What sport teams have you played in? *

- ☐ Football
- ☐ Basketball
- ☐ Ice Hockey
- ☐ Field Hockey
- ☐ Cheerleading
- ☐ Swimming
- ☐ Wrestling
- ☐ Soccer
- ☐ Volleyball
- ☐ Indoor Track
- ☐ Outdoor Track
- ☐ Cross Country
- ☐ Baseball
- ☐ Softball
- ☐ Tennis
- ☐ Lacrosse
- ☐ Ski
- ☐ Golf
- ☐ None
- ☐ Other...

We wanted to see if there are certain sports associated with significantly higher rates of athlete depression or anxiety than others.

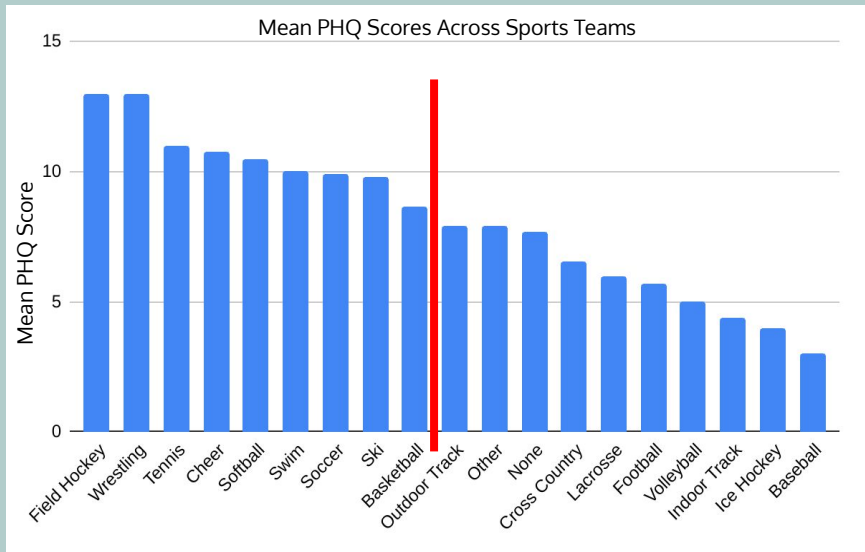
If our results show that students who play certain sports tend to have significantly worse depression or anxiety, it could indicate something about that sport in particular.

Note: Non-WRHS sports (which included mountain biking, snowboarding, equestrian, flag football, gymnastics, dance, etc) were grouped under "Other"

PHQ Score	Baseball	Basketball	Cheerleading	Soccer	Outdoor Track	Cross Country	Softball	Other	Field Hockey	Wrestling	Ski	Cheer	Lacrosse	Football	Swim	Ice Hockey	Indoor Track	None	Tennis	Volleyball	
PHQ Score	9	3	18	18	18	18	18	18	18	18	18	16	7	4	14	7	4	0	6	27	4
PHQ Score	3	11		1	18	4	12	6	10	8	12	6	8		20	7		11	1	0	6
PHQ Score	0	18		16	4	6	4	0	27		7	12			0	16		0	8	6	0
PHQ Score	3	18		6	12	3	7	14	12		5				0			3	16		11
PHQ Score	0	4		12	5	0	18	9	4		9				3			8	5		4
PHQ Score		1		4	0	11	4	0	7						4				8		
PHQ Score		16		21	4	4		2							7				3		
PHQ Score		3		5	11			2							4				4		
PHQ Score		4		0	3			9							0				16		
PHQ Score				1	8			19							5				3		
PHQ Score				11	4														11		
PHQ Score				6															12		
PHQ Score				2															18		
PHQ Score				13															1		
PHQ Score				19															3		
PHQ Score				12																	
PHQ Score				3																	
PHQ Score				8																	
PHQ Score				19																	
PHQ Score				27																	
PHQ Score				4																	
Mean	3	8.66666667	18	9.904761905	7.909090909	6.571428571	10.5	7.9	13	13	9.8	8.333333333		6	5.7	10	4	4.4	7.666666667	11	5
St. Dev	3.674234614	7.071067812	#DIV/0!	7.738893732	6.090230627	6.051367417	6.503845017	7.171703656	8.342661446	7.071067812	4.324349662	3.214550254	2.828427125	6.549809158	5.196152423		0	4.929503018	5.69042638	14.17744688	4

Nine sports had an average PHQ score higher than the average for WRHS, ten had an average PHQ score lower than the average for WRHS.

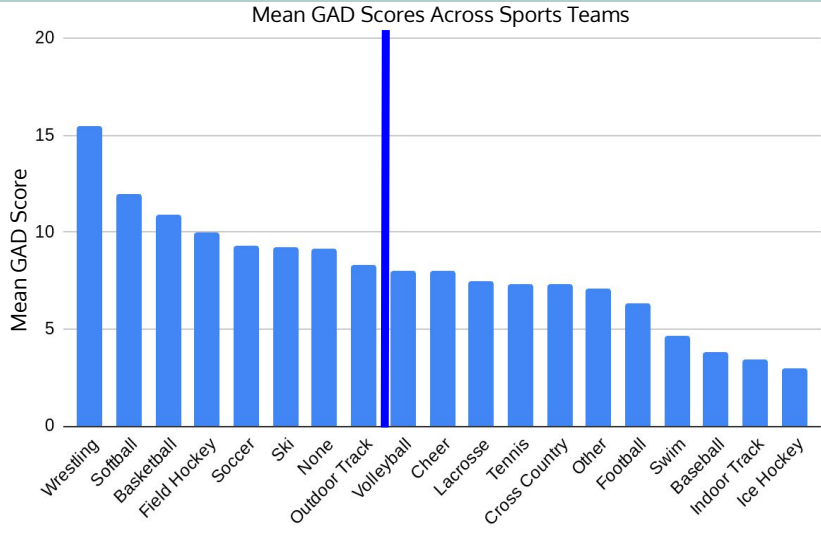
SPORTS: DEPRESSION



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	
1	GAD7 Score	Baseball	Basketball	Cheerleading	Soccer	Outdoor Track	Cross Country	Softball	Other	Field Hockey	Wrestling	Ski	Cheer	Lacrosse	Football	Swim	Ice Hockey	Indoor Track	None	Tennis	Volleyball	
2	GAD7 Score		9	3	15	15	15	15	15	19	19	18	6	3	12	6	3	0	13	16	6	
3	GAD7 Score		4	14		11	19	8	18	3	3	12	8	3	12	20	4	5	17	1	1	
4	GAD7 Score		0	15		18	8	3	8	0	21		6	8		1	4	0	3	5	11	
5	GAD7 Score		5	19		3	8	5	3	10	8		6			6		1	16		10	
6	GAD7 Score		1	8		8	6	0	16	9	3		8			4		11	2		12	
7	GAD7 Score			11		6	0	8	12	2	6					6			12			
8	GAD7 Score			18		5	3	12		6						3			6			
9	GAD7 Score			6		6	8			5						0			8			
10	GAD7 Score			4		8	1			8						5			17			
11	GAD7 Score					0	11			13						6			1			
12	GAD7 Score					16	12												7			
13	GAD7 Score					10													12			
14	GAD7 Score					4													18			
15	GAD7 Score					10													0			
16	GAD7 Score					15													5			
17	GAD7 Score					13																
18	GAD7 Score					1																
19	GAD7 Score					11																
20	GAD7 Score					13																
21	GAD7 Score					16																
22	GAD7 Score					6																
23	Mean	3.8	10.8888889		15	9.285714286	8.272727273	7.285714286	12	7.1	10	15.5	9.2	5.666666667	7.5	6.3	4.666666667	3	3.4	9.133333333	7.333333333	8
24	St. Dev	3.563705936	5.967504597	#DIV/0!	5.207137958	5.798118817	5.154748158	5.621387729	4.817791103		8	4.949747468	5.019960159	2.516611478	6.363961031	5.831904587	1.154700538	0	4.722287581	6.277700519	7.767453465	4.527692569

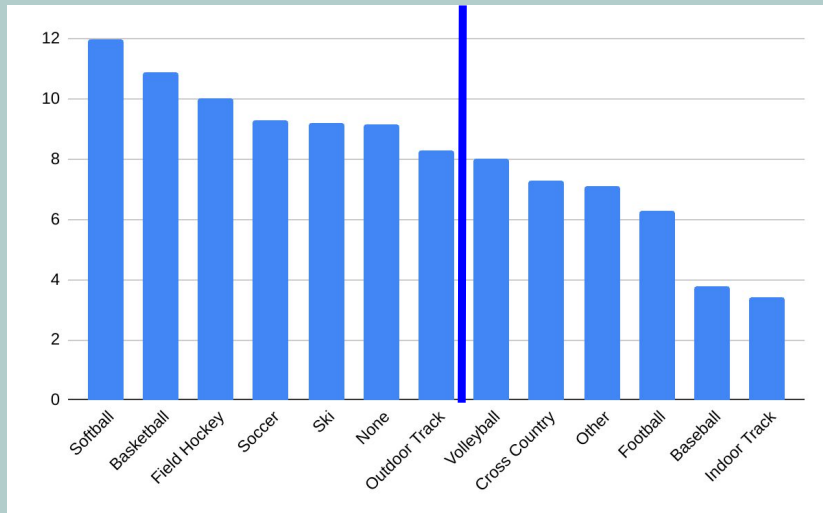
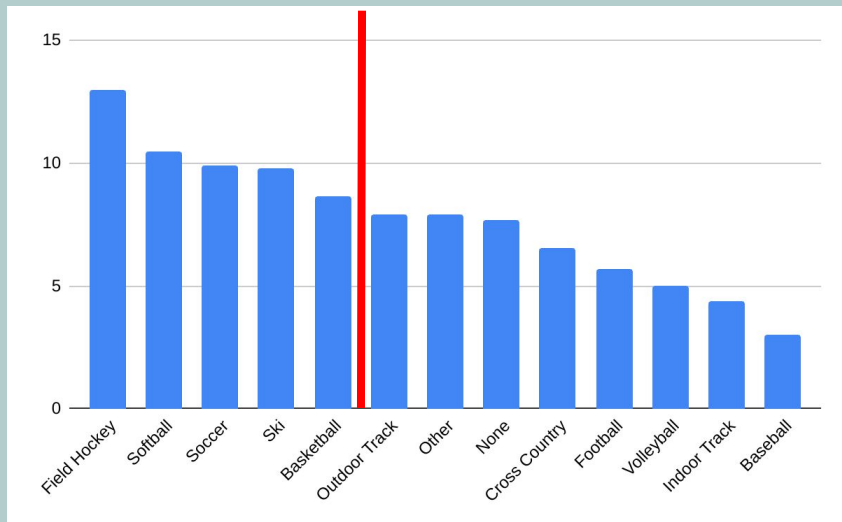
Eight sports had an average GAD score higher than the average for WRHS, eleven had an average GAD score lower than the average for WRHS.

SPORTS: ANXIETY



To reduce error caused by a small sample size, these graphs only contain data for sports that had ≥ 5 respondents.

These graphs are more accurate but only speak for the more popular sports.



What music genres do you regularly listen to? *

☐ Hip-Hop or Rap

☐ Pop

☐ Rock

☐ Punk or Metal

☐ Country or Folk

☐ Classical, Jazz, or Lo-Fi

☐ Blues, R&B, or Soul

☐ Unconventional (East Asian pop music, anything with the -core suffix, etc.)

☐ I do not listen to music.

☐ Other...

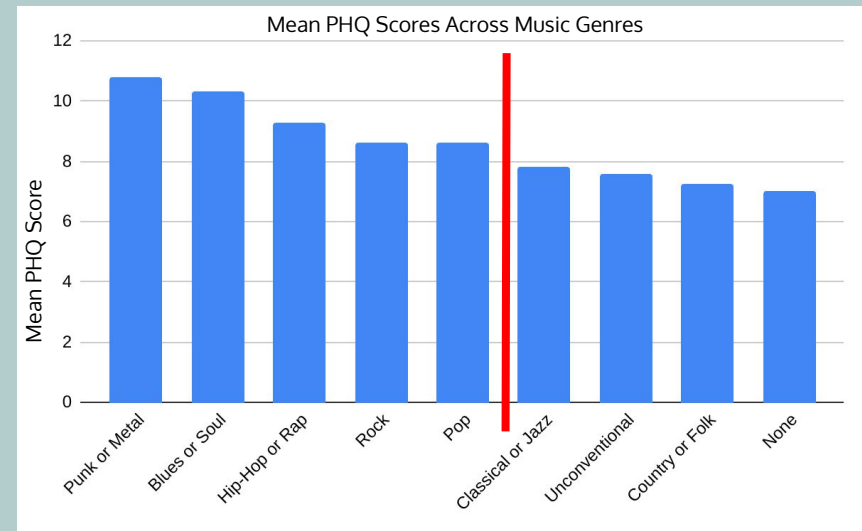
MUSIC GENRE QUESTION

We wanted to see if there are certain music genres associated with significantly higher rates of depression or anxiety than other genres.

Note: Custom responses to the music genre question (which included Worship music, audioslave, Latin music, etc) were grouped under "Unconventional"

It's ready the PHQ.

L	M	N	O	P	Q	R	S	T	U
GAD Score	Classical or Jazz	Country or Folk	Hip-Hop or Rap	Blues or Soul	Unconventional	Pop	Rock	Punk or Metal	None
GAD Score	3	4	0	27	0	3	12	1	7
GAD Score	18	0	14	18	6	11	4	8	
GAD Score	0	4	0	0	16	1	1	21	
GAD Score	16	19	12	11	11	9	8	13	
GAD Score	4	11	27	14	7	4	21	4	
GAD Score	5	16	18	3	3	18	13	16	
GAD Score	11	9	0	6	0	8	4	20	
GAD Score	4	12	4	16	4	11	16	11	
GAD Score	19	4	3	16	19	14	5	7	
GAD Score	4	4	11	9	5	3	3	8	
GAD Score	2	3	1	13	18	6	0	10	
GAD Score		0	9	4	2	16	20		
GAD Score		6	4	11	4	19	11		
GAD Score		0	18	11	3	11	7		
GAD Score		3	8	0	11	16	3		
GAD Score		8	11	6	12	9	6		
GAD Score		6	14			12	16		
GAD Score		27	3			4	2		
GAD Score		18	6			1	18		
GAD Score		11	16			8	2		
GAD Score		1	19			21	8		
GAD Score		7	11			13	10		
GAD Score		6	16			4			
GAD Score		4	9			16			
GAD Score		2	12			9			
GAD Score		5	4			0			
GAD Score		16	1			1			
GAD Score		2	8			11			
GAD Score		2	21			0			
GAD Score			13			4			
GAD Score			4			19			
GAD Score			16			6			
GAD Score			5			0			
GAD Score			3			3			
GAD Score			0			8			
GAD Score			20			6			
GAD Score			11			27			
GAD Score			7			18			
GAD Score			3			11			
GAD Score						1			
GAD Score						7			
GAD Score						6			
GAD Score						4			
GAD Score						2			
GAD Score						5			
GAD Score						3			
GAD Score						6			
GAD Score						16			
GAD Score						2			
GAD Score						18			
Mean	7.818181818	7.24137931	9.282051282	10.3125	7.5625	8.62	8.636363636	10.81818182	7
St. Dev	6.896639498	6.626904896	6.924205934	7.182110665	6.196437148	6.552425941	6.425732899	6.274044657	0

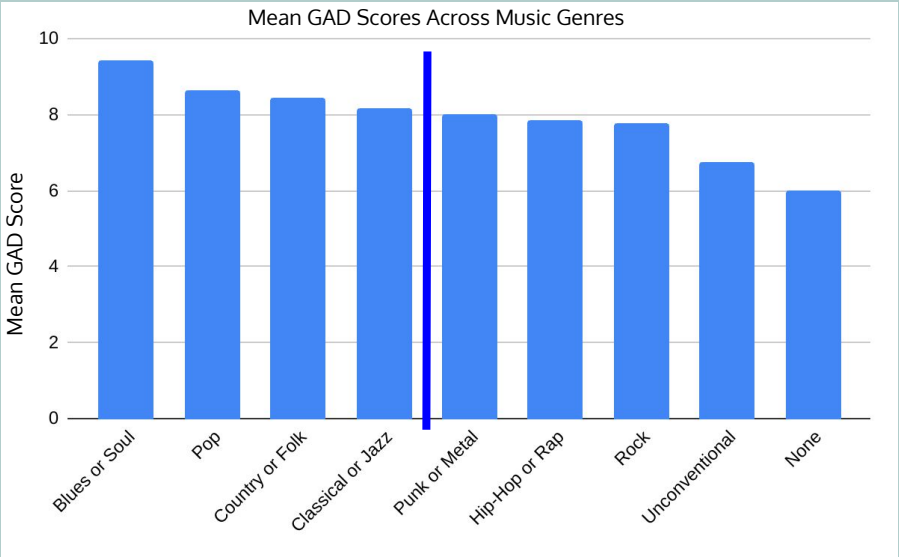


Five categories had an average PHQ score higher than the average for WRHS, four had an average PHQ score lower than the average for WRHS.

MUSIC: DEPRESSION

	A	B	C	D	E	F	G	H	I	J
1	GAD Score	Classical or Jazz	Country or Folk	Hip-Hop or Rap	Blues or Soul	Unconventional	Pop	Rock	Punk or Metal	None
2	GAD Score	6	4	0	16	0	1	8	17	6
3	GAD Score	15	11	12	15	1	5	8	3	
4	GAD Score	0	3	6	0	4	0	17	5	
5	GAD Score	16	13	13	16	7	9	3	10	
6	GAD Score	3	8	16	10	4	3	5	3	
7	GAD Score	2	17	15	3	5	19	10	4	
8	GAD Score	7	9	0	1	8	11	3	20	
9	GAD Score	8	8	3	16	8	16	4	7	
10	GAD Score	15	8	1	17	15	10	2	4	
11	GAD Score	12	3	5	9	6	3	4	12	
12	GAD Score	6	4	0	10	18	1	1	3	
13	GAD Score		1	9	3	5	16	20		
14	GAD Score		13	3	7	6	13	7		
15	GAD Score		1	19	10	1	8	4		
16	GAD Score		3	11	8	8	17	5		
17	GAD Score		12	16	10	12	9	10		
18	GAD Score		3	10			8	18		
19	GAD Score		21	3			8	4		
20	GAD Score		16	1			17	18		
21	GAD Score		14	16			3	5		
22	GAD Score		11	13			5	12		
23	GAD Score		6	8			10	3		
24	GAD Score		5	17			3			
25	GAD Score		12	9			4			
26	GAD Score		6	8			7			
27	GAD Score		6	8			2			
28	GAD Score		18	17			0			
29	GAD Score		4	3			10			
30	GAD Score		5	5			8			
31	GAD Score			10			8			
32	GAD Score			3			15			
33	GAD Score			4			13			
34	GAD Score			2			1			
35	GAD Score			4			3			
36	GAD Score			1			12			
37	GAD Score			20			3			
38	GAD Score			7			21			
39	GAD Score			4			16			
40	GAD Score			5			14			
41	GAD Score						11			
42	GAD Score						6			
43	GAD Score						5			
44	GAD Score						12			
45	GAD Score						6			
46	GAD Score						6			
47	GAD Score						5			
48	GAD Score						10			
49	GAD Score						18			
50	GAD Score						4			
51	GAD Score						18			
52	Mean	8.181818182	8.448275862	7.871794872	9.4375	6.75	8.66	7.772727273	8	6
53	St. Dev	5.582442001	5.41549501	5.92132403	5.587246347	4.505778805	5.855447076	5.77294031	6.016643583	0

MUSIC: ANXIETY



Five categories had an average GAD score higher than the average for WRHS, five had an average GAD score lower than the average for WRHS.

Improvements for Future

- Obtain a larger sample size (instead of only selecting 200 students select 400). This is because the number of respondents/answers was limited
- More questions to ask
- Be more specific about the "Sports" question - confusion with other extracurriculars